

Final Project



**University
of Victoria**

MECH 493 - Design of Thermo-Fluid Systems

Prepared for:

Dr. Samira Gharehkhani

Group 3

Moyo Adeoye

Will Fox

Erick Soderquist

Connor Johnston

Luca Carinelli

Due: April 4, 2026

Table of Contents

Summary.....	1
1. Project Description.....	1
1.1. Motivation.....	1
1.2. Objectives and Project Scope.....	2
2. Technical Report.....	4
2.1. Overview of Methodology.....	4
2.2. Detailed Description of Methods.....	5
2.3. Key Design Decisions.....	6
2.4. Results.....	8
3. Budget Report.....	11
3.1. Design implementation budget.....	11
3.2. Work Schedule.....	14
4. Concerns and Issues.....	14
5. Recommendations.....	15
6. Conclusions.....	15
7. References.....	17
Appendix A: Original System Calculations.....	18
Appendix B: Proposed System Design Calculations.....	22

List of Tables and Figures

Figure 01: Apparatus, labelled.....	2
Figure 02: System schematic.....	4
Figure 03: UV Sterilizer System Curve.....	6
Figure 04: System Schematic Block Diagram.....	8
Figure 05: Front View of Optimized Design.....	9
Figure 06: Isometric View of Optimized Design.....	9
Figure 07: Energy consumed versus gallons produced in each design.....	11
Figure A1: Original system assumptions.....	18
Figure A2: Original system calculations for all five operating cases 1/2.....	19
Figure A3: Original system calculations for all five operating cases 2/2.....	20
Figure A4: Minor loss coefficients used in calculations [1].....	20
Figure A5: Absolute roughness values used in calculations [1].....	21
Figure A6: Moody chart used in calculations [1].....	21
Figure B1: Optimized system design calculations 1/2.....	22
Figure B2: Optimized system design calculations 2/2.....	23
Table 01: Operating cases of current filtration device.....	3
Table 02: Design decision summary.....	8
Table 03: System head loss contribution summary.....	10
Table 04: Original system BOM.....	12
Table 05: Proposed design BOM.....	12
Table 06: Engineering budget report.....	13
Table 07: Updated work schedule.....	14

Summary

This project evaluated and redesigned a laboratory water filtration system to improve efficiency, reduce energy consumption, and lower manufacturing cost while maintaining ultra-pure water output. A hydraulic analysis was performed using the energy equation and Darcy–Weisbach relations to calculate frictional and minor losses for five possible operating configurations. Based on the results, Case 5 (faucet supplied inlet only) was selected as the optimal configuration, as it provided sufficient pressure without requiring a pump.

The redesigned system removes redundant components, including the reverse osmosis loop, storage tank, and pump, and instead uses a simplified arrangement consisting of a cartridge filter, UV sterilizer, and two deionization tanks in series. The optimized layout reduces pipe length and fittings, lowering total head loss by 20.5% and decreasing energy consumption to approximately 33% of the original system requirement. Manufacturing cost was also reduced by 56.2%, resulting in a more compact, cost-effective, and energy-efficient design suitable for laboratory applications.

1. Project Description

1.1. Motivation

Water filtration systems are used to produce clean, high purity water for several purposes such as drinking, laboratory experiments, and critical manufacturing processes like pharmaceuticals and electronics. These systems typically include several filtration stages that can include reverse osmosis (RO), sediment filtration, UV sterilization, and deionization. While these filters are robust for producing ultra-pure water, there are often substantial hydraulic losses and high manufacturing costs associated with the system.

The current water filtration device in the mechanical engineering department was procured as a legacy design for a different application unrelated to fluid mechanics experiments. This system contains several pipe fittings and oversized components that are not essential for the application, introducing hydraulic losses, increasing operational energy consumption, and occupying a large spatial volume. Thus, the motivation for this project stems from a need for a purpose-built water filtration device for fluid mechanics experiments. The current apparatus is shown below in figure 01:

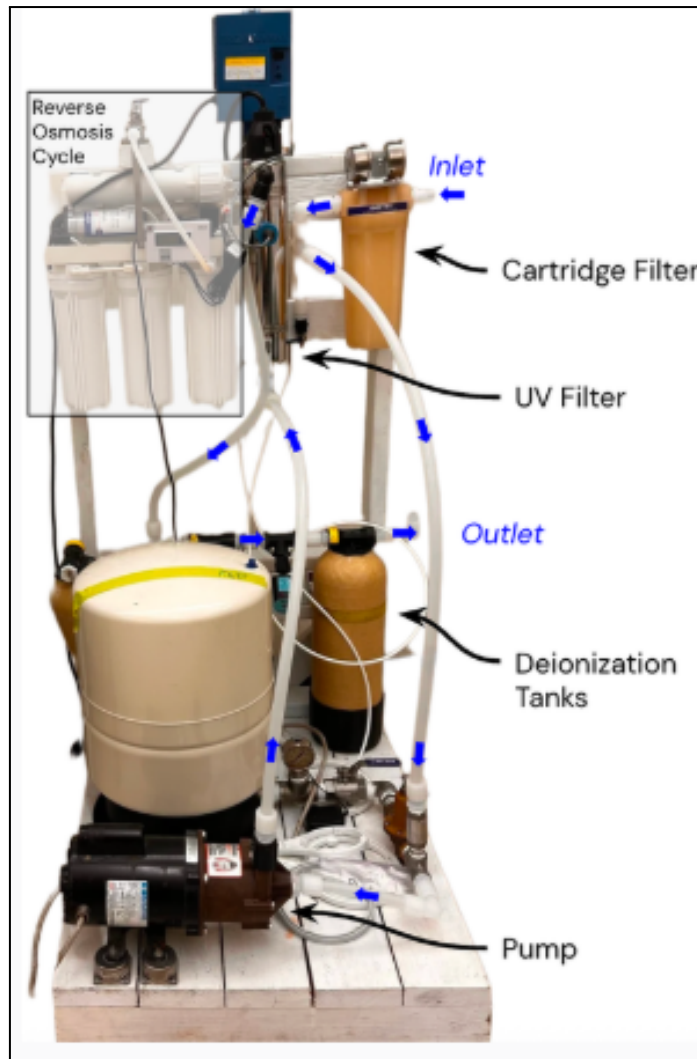


Figure 01: Apparatus, labelled

1.2. Objectives and Project Scope

The primary objective of this project is to quantify the required pump head across five discrete operating cases for the current dual-inlet water filtration device to determine a hydraulic baseline with existing hardware. The system has two distinct inlets connected at a junction tee at the pump inlet. One inlet branch, branch “A”, is an RO filtration process to a pressurized holding tank, whereas the other inlet branch, branch “B”, has a sediment cartridge filter that can be either from pumped stillwater or from a pressurized source like a laboratory faucet. The following table summarizes the operating cases to be analyzed:

Table 01: Operating cases of current filtration device

Case ID	Branch A inlet	Branch B inlet
1	Active	Pumped stillwater
2	Active	Faucet supplied
3	Null	Pumped stillwater
4	Active	Null
5	Null	Faucet supplied

The study then looks to identify the most suitable operating condition for the laboratory application. Once the optimal case is selected, the project transitions into a redesign phase focused primarily on removing design redundancies, minimizing head losses, and reducing the manufacturing cost to produce multiple units for expanding access to other lab rooms. Moreover, the redesign will attempt to reduce the device's spatial volume and minimize the energy consumption between the pump and the UV sterilizer. Completing the prescribed objectives will provide a simplified system architecture that is cost-effective, operationally efficient, easy to maintain, and provides ultra-pure water for laboratory experiments.

The scope of this project includes hydraulic modelling, design optimization, and energy analysis of the current water filtration system. Initially, the project involves characterizing the hydraulic parameters and required pump head, if any, of the existing system across each of the five operating cases. In the operating case where the pressurized tank in branch A is active, a flow rate of 0.5 GPM at 35 PSIG is assumed from the product specification. In the case of a supply faucet in branch B, a flow rate of 3 GPM at 50 PSIG is assumed, whereas the stillwater source has an unknown flow rate at an atmospheric pressure of 0 PSIG. The pump and filters are rated for approximately 30 GPM, thus, finding the duty point from the manufacturers specification sheet will help determine the actual flow through the system

The scope also includes selecting the most suitable application scenario for the system based on the data from the hydraulic models, and trying to quantify optimization opportunities that include reducing pipe lengths, selecting pressure rated smooth pipe materials, creating an optimized CAD model layout in SolidWorks, and reducing the number of fittings required for functionality. Investigating redundancies in the current system's

dual-inlet filtration process to streamline the process and minimize losses is also in the scope of the project. Once the final design is completed, an energy consumption study will be carried out to compare the amount of electrical energy required to produce a prescribed quantity of filtered water between the current system and the proposed system. The hydraulic calculations are based off of the system schematic outlined in the figure below:

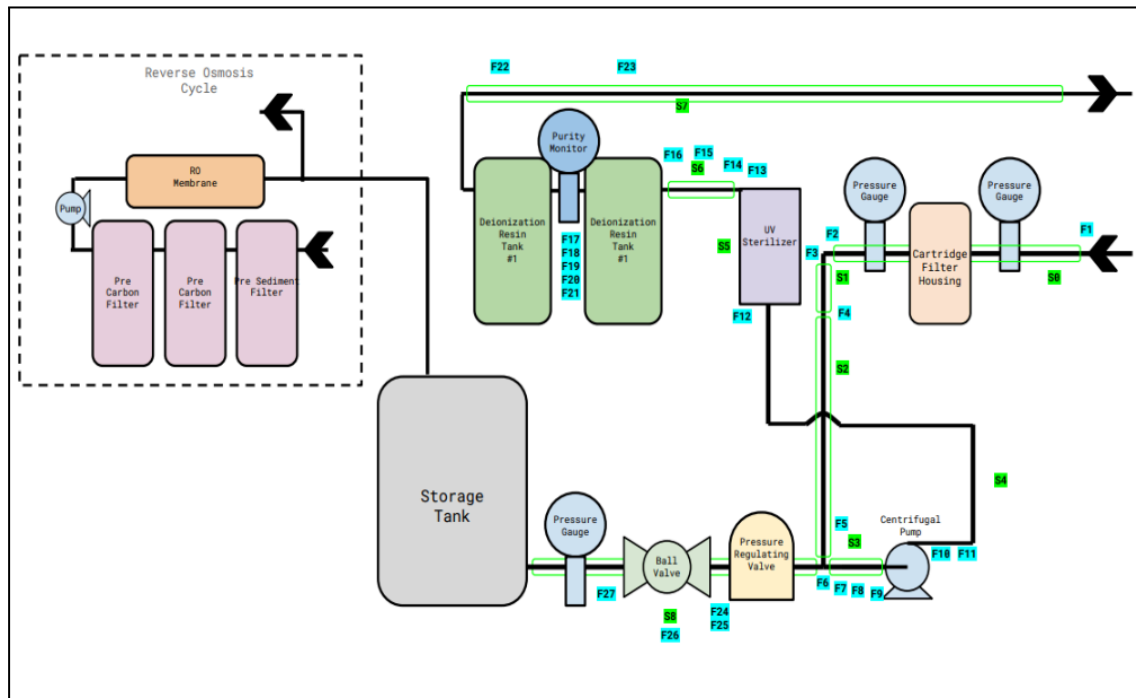


Figure 02: System schematic

Case 5 (faucet supplied inlet only) was selected as the preferred operating condition because the laboratory faucet provides sufficient pressure to drive the filtration process without requiring a pump. Eliminating the pump and RO loop reduces system complexity, energy consumption, and cost while maintaining adequate filtration performance for laboratory applications.

2. Technical Report

2.1. Overview of Methodology

The performance of the water filtration system can be quantified by applying the principle of conservation of energy between the inlet and outlet of the system. The primary objective of this methodology is to determine the total energy required to move the fluid from the source to the outlet while overcoming friction losses and losses from the different filters and fittings. Modelling the required pump head and total system head losses will allow us to

determine if the laboratory faucet provides sufficient pressure to drive the flow. In the energy analysis, frictional losses from the pipe roughness and minor losses from the fittings and major components are incorporated using empirical data to obtain an accurate representation of the required pump head.

2.2. Detailed Description of Methods

The performance of the water filtration system was analyzed using the energy balance equation over the entire system. Between the inlet (point 1), and the outlet (point 2), the energy equation with incorporated losses is as follows:

$$\frac{P_1}{\rho g} + \frac{V_1^2}{2g} + z_1 + h_p = \frac{P_2}{\rho g} + \frac{V_2^2}{2g} + z_2 + h_L \quad (1)$$

Where P is pressure, V is velocity, ρ is the fluid density, z is the elevation rise to the prescribed datum, h_p is the required pump head to move the fluid from point 1 to point 2, and h_L is the incurred losses from frictional losses and minor losses, shown below in equation (2):

$$h_L = h_f + h_m \quad (2)$$

The system was broken down into the individual pipe segments and components so that we could calculate the head losses step by step through the flow path. Frictional losses in the pipe segments were calculated using the Darcy-Weisbach equation (3):

$$h_f = f \frac{L}{D} \frac{V^2}{2g} \quad (3)$$

Where h_f is the friction head loss, f is the friction factor, L is the pipe length, D is the pipe diameter, and V is the flow velocity. Minor losses from fittings like elbows, tees, valves, and reducers were calculated using standard loss coefficients, or K values, obtained from empirical data in [1]. The minor losses for the system are then represented by equation (4):

$$h_m = \left(\sum_{i=1}^n K_i \right) \frac{V^2}{2g} \quad (4)$$

Where h_m is the minor loss and K_i is the loss coefficient for each fitting. The system was divided into eight pipe segments, and all fittings were assigned to their respective segments. For the main sediment filter where a minor loss coefficient was not readily available from the manufacturer, empirical data suggests that an average pressure differential of approximately 2 psi across the filter can be expected for filters in this flow class before they are saturated [2]. Knowing the pressure drop across the component, equation (5) is used to compute the minor headloss from the component :

$$h_L = \frac{\Delta P_{1,2}}{\rho g} \quad (5)$$

Similarly for the deionization tanks, the pressure differential in this flow class can be assumed to be approximately 5 psi in optimal working condition according to [3], where the minor head loss can be computed for the energy equation.

The pressure drop across the UV sterilizer is directly proportional to the flow through the system and this was identified using a system curve from a model with similarly rated flow and construction[3]. The curve is shown in Figure 03 below:

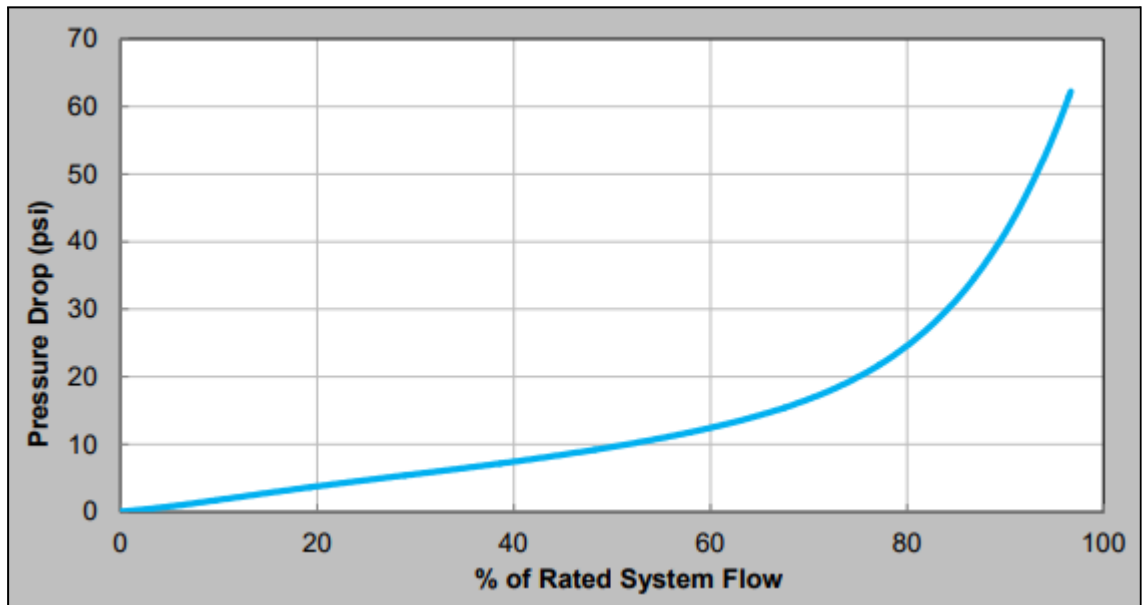


Figure 03: UV Sterilizer System Curve

The total pressure head was calculated as a function of the flow rate to provide a system curve. The system curve was subsequently compared with the manufacturer's pump curve data to determine the actual duty point for improved comprehension of the systems behaviour without physical testing available.

2.3. Key Design Decisions

The redesign of the water filtration system involved multiple critical engineering choices for reducing complexity, redundancy, lowering lifecycle costs, and ensuring the highest possible water purity for laboratory experiments. These decisions were made evaluating the hydraulic model and schematic made for the original design and comparing them to requirements needed for retrofitting the system for the lab.

The primary change to the system was the removal of the reverse osmosis (RO) filtration stages. The current system uses a dual-inlet filtration process, where the two discrete filtration pipelines connect at a junction tee near the pump inlet. This results in

non-uniformly filtered water since the sediment filter on one inlet does not provide the same level of purity or remove the same particles as an RO does. It was found that the RO was likely used as a pretreatment to extend the lifespan of the deionizer resins, removing salts, bacteria, and impurities for an effective purity up to 99% [4]. The deionizers produce ultra-pure water with close to zero Total Dissolved Solids (TDS) remaining by filtering out any ions that an RO system cannot, making it a common application for engineering labs and pharmaceutical labs, with the exception that these filters do not remove bacterial growth or viruses [4]. Thus, with a sediment filter pretreatment for minerals and impurities, an ultra-violet filter treatment for bacteria and viruses, the RO system becomes redundant since it does not produce pure enough water for critical laboratory applications compared to a DI process. This change effectively allows for a single inlet filtration process to reduce redundancy and ensure that the water is uniformly filtered.

In addition to removing the RO system, a central goal for the optimization of the system is adapting it for the lab room specifically, where a faucet supply can be used in place of a centrifugal pump. If the designed system is computed to have sufficient head pressure available from the faucet, manufacturing costs and operational costs related to energy consumption can be drastically reduced without the need for an explicit pump. This transition, although beneficial for incurred project costs, will inherently reduce the flow capacity and make the process of obtaining filtered water from the system substantially slower.

Another key decision made was to keep the deionizer tanks in series rather than parallel, even though the leading tank will saturate first. According to [3], putting these tanks in series adds beneficial redundancy in the lagging tank to capture impurities missed by the leading tank. To solve the issue of the leading tank saturating first, the proposed design must be designed for ease of maintenance, where the two tanks can be swapped in order once the first tank is close to saturation. At the expense of ultra-purity, the tanks cannot produce high volumetric flow like parallel tanks can; however, the flow rate is a limiting factor from the faucet and the selected tanks have the individual capacity for the entire supplied flow rate.

Lastly, to ensure the lab is using near ultra-pure water, it is critical to remove the holding tank and make the filtration process work only on demand. The reason for this is that a holding tank can promote microbial growth and will nullify the validity of experimental data collected in the lab [5]. The main design decisions are summarized in the following table:

Table 02: Design decision summary

Design Decision	Justification
Removal of RO system	Reduce filtration redundancy; improve uniformity of filtered water; reduce cost
Removal of centrifugal pump	Designing around the lab faucet reduces manufacturing and energy consumption costs; removes a potential point of required maintenance
Series deionizing tanks	Adds redundancy in the case of one tank failing; produces purer water; ideal for lower flow rates like the proposed system.
Removal of holding tank	Tank inhibits ultra-pure water by promoting microorganism growth

2.4. Results

After the analysis was completed and redundancies from the previous system were removed, the final system design has the cartridge filter height lowered and connected through a short pipe run and an elbow fitting to the inlet source above it, the UV sterilizer outlet and the deionization tank inlets at the same height. All pipe runs between filters were kept to a minimum to reduce losses. A block diagram of the proposed system schematic is shown in Figure 04 below and detailed CAD models are shown in Figure 05 and Figure 06.

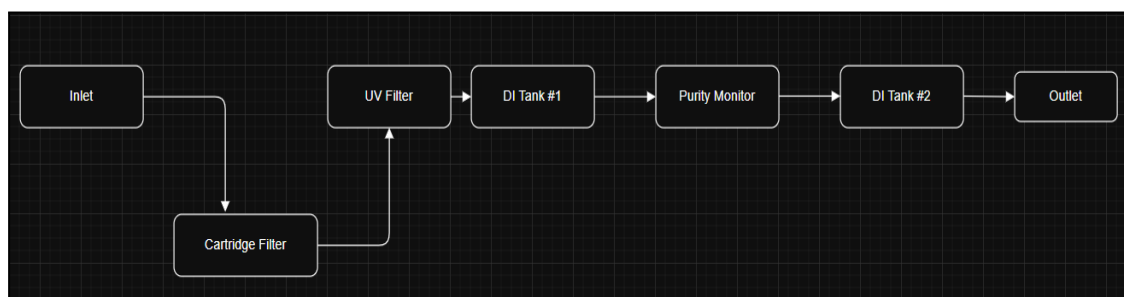


Figure 04: System Schematic Block Diagram

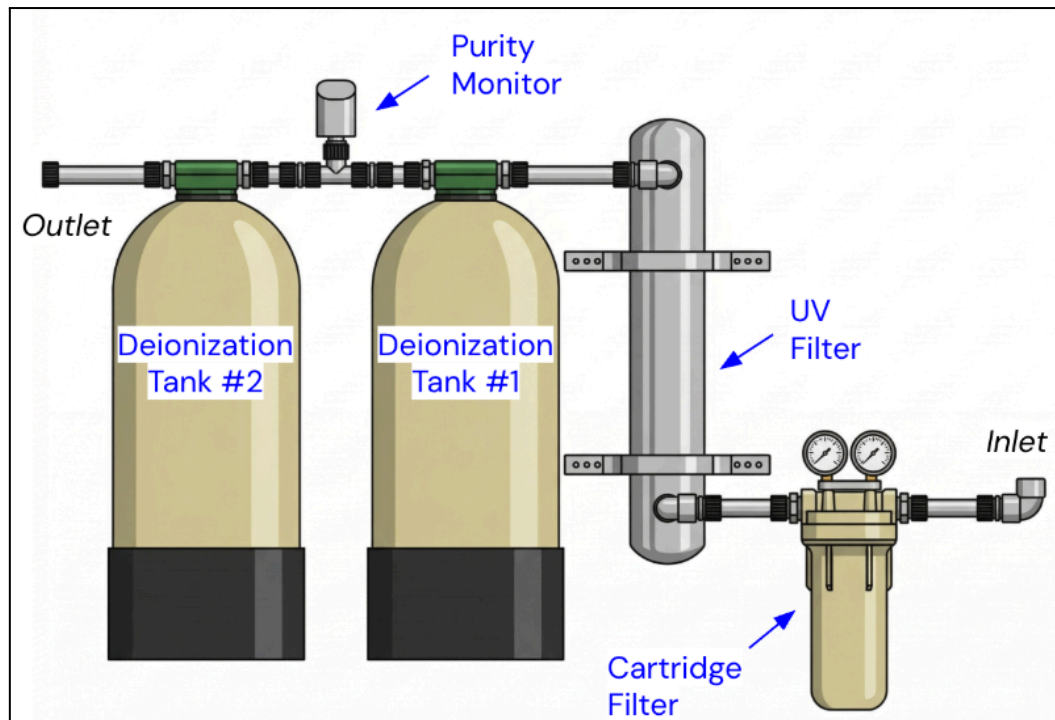


Figure 05: Front View of Optimized Design

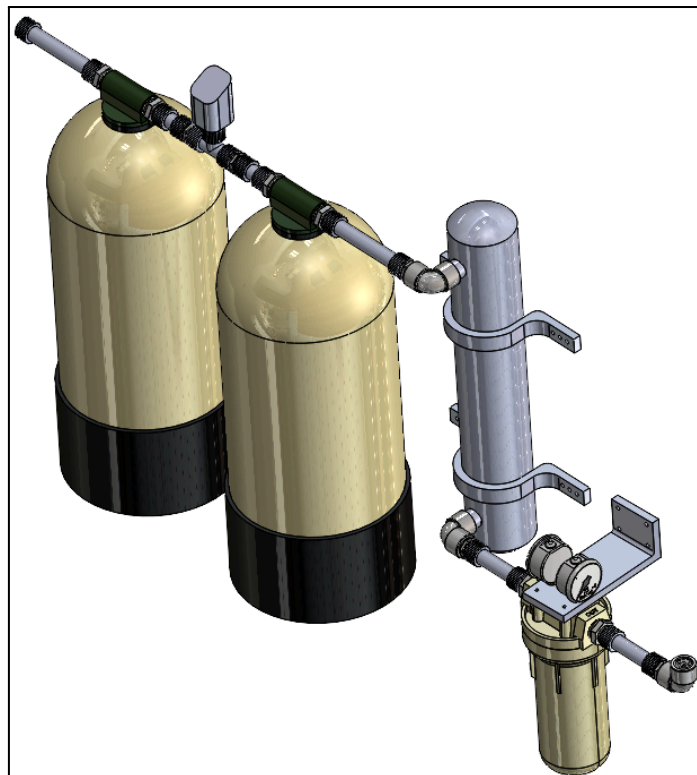


Figure 06: Isometric View of Optimized Design

The proposed system significantly reduced pipe segment lengths, leading to a large reduction in friction head loss contributions up to 90.32%. Similarly, minor head losses in the system were reduced by simplifying the overall layout and minimizing the amount of fittings; however, the minor loss savings were only marginally reduced compared to friction savings due to significant pressure drop contributions from the main filters. The minor head loss was reduced by up to 18.22%, and the overall weighted reduction of losses was found to be 20.51%. Reducing total head loss decreases the required driving pressure, enabling the system to operate using the available faucet pressure without requiring additional pumping power. The results are summarized in the following table:

Table 03: System head loss contribution summary

	Legacy system (case 5)	Proposed system (case 5)	Percent reduction
h_{fric}	0.6375m	0.0617m	90.32%
h_{min}	19.8041m	16.1950m	18.22%
$h_{L,to}$	20.44m	16.257m	20.51%

As the two systems outlined in the report do not produce the same flow rate, it is important to compare their respective energy consumption rates from the pump and ultraviolet filter per capacity of produced purified water. According to manufacturers data for each component, the original system's UV filter was found to require 230W to sterilize 30 gallons of water in series with a pump consuming approximately 825W of continuous duty power. In the optimized system, the UV requires 120W to sterilize 10 gallons of water. The energy consumption for each design can be modelled over time in Watt-hours compared to the amount of purified water produced, as shown in the following figure:

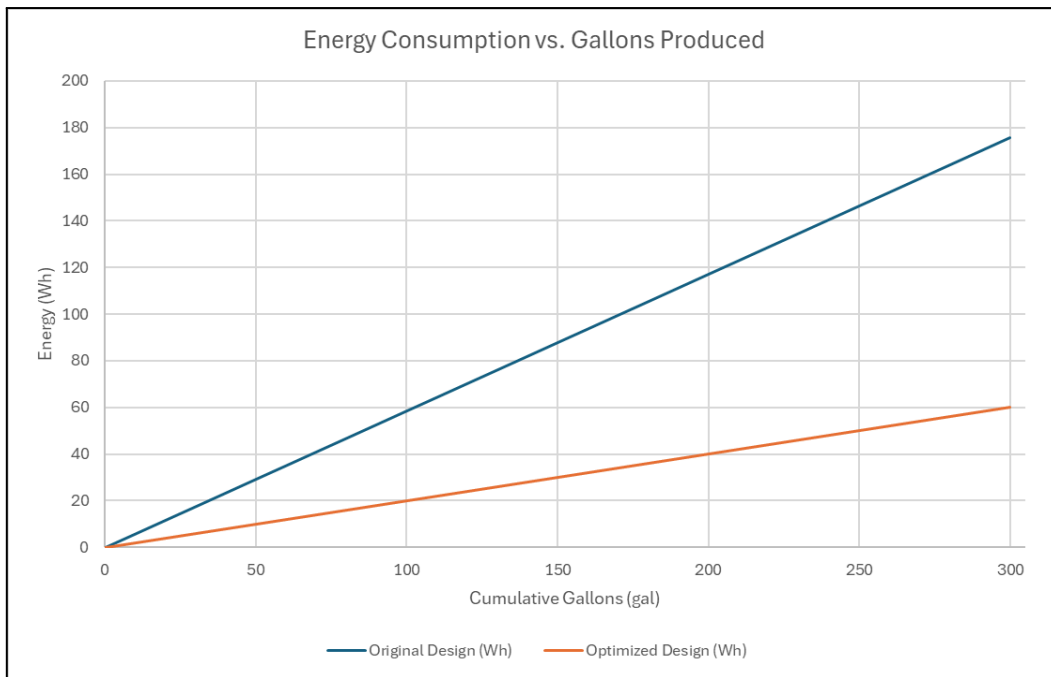


Figure 07: Energy consumed versus gallons produced in each design

The energy consumption plot shows that producing a standard 1000 litre IBC tank worth of purified water for laboratory use with the proposed design only consumes approximately 33% of the energy required to produce the same capacity with the original design. Since the rate of consumption in the proposed system was found to be lower, over prolonged usage of the system, the energy savings can become significant.

3. Budget Report

3.1. Design implementation budget

Through further investigation for each individual component in the current system, manufacturing a second unit without design or labour consideration would cost an estimated \$11,440.52 CAD using reputable suppliers with readily available components. The proposed design used smaller capacity filters to function better with the provided flow rate from the laboratory faucet, and overall required less pipe fittings. The estimated cost to produce a single unit was found to be \$5014.00 CAD, representing a 56.17% decrease. The complete bill of materials for the original design and proposed design are summarized below in the following figures:

Table 04: Original system BOM

Pump Filtration System (BOM)						
ID	Category	Part Description	Supplier	Orig Qty	Unit Cost (\$ CAD)	Original Total (\$ CAD)
1	Piping / Fittings	Pipe nipple, 4", 316 SS	McMaster-Carr	1	\$22.94	\$22.94
2	Piping / Fittings	Tee, 1" FNPT, 316 SS	McMaster-Carr	1	\$27.44	\$27.44
3	Instrumentation	Pressure Gauge, 1/4"	McMaster-Carr	3	\$22.71	\$68.14
4	Piping / Fittings	pipe nipple, 1", close, 316 SS	McMaster-Carr	1	\$13.61	\$13.61
5	Valve	Ball valve, 1"	McMaster-Carr	1	\$85.20	\$85.20
6	Piping / Fittings	Bushing, 1" MNPT x 1/4" FNPT, 304 SS	McMaster-Carr	1	\$13.90	\$13.90
7	Piping / Fittings	Pipe elbow, 1" MNPT x 1" FNPT, 316 SS	McMaster-Carr	1	\$35.44	\$35.44
8	Piping / Fittings	Bushing, 1" MNPT x 0.75" FNPT, 304 SS	McMaster-Carr	1	\$12.27	\$12.27
9	Piping / Fittings	pipe nipple, close, 0.75"	McMaster-Carr	3	\$4.90	\$14.69
10	Valve	Pressure Regulating Valve, 25-75PSI	Home depot	1	\$199.00	\$199.00
11	Piping / Fittings	Tee, 0.75" FNPT, 316 SS	McMaster-Carr	1	\$18.99	\$18.99
12	Valve	Check Valve, 316SS, 0.75" FNPT	McMaster-Carr	1	\$92.19	\$92.19
13	Piping / Fittings	Pipe elbow, 0.75" MNPT, Polypropylene	McMaster-Carr	4	\$6.28	\$25.13
14	Piping / Fittings	Pipe elbow, 0.75" FNPT, PVC	McMaster-Carr	2	\$2.24	\$4.49
15	Pump	March Centrifugal Pump, 1/2Hp, 60Hz,	BMP Industrial Supply	1	\$1,155.00	\$1,155.00
16	Treatment	UV Sterilizer, Trojan UV Max pro30, 30GPM	esp Water	1	\$3,642.83	\$7,412.00
17	Piping / Fittings	Tee, 0.75" FNPT, PVC	McMaster-Carr	2	\$4.52	\$9.03
18	Treatment	Purity monitor, hydro-check systems	Hydro-check	1	\$95.00	\$95.00
19	Treatment	Deionization Resin Tank	Nationwide DI	2	\$479.50	\$959.00
20	Filtration	RO Filtration System	Home depot	1	\$358.00	\$358.00
21	Pump	15GPH transfer pump	Amazon	1	\$20.00	\$20.00
22	Filtration	Ametek sediment filter	Grainger	1	\$142.46	\$142.46
23	Tank	Storage tank, 22 Gal, 3/4 NPT	McMaster	1	\$588.05	\$588.05
24	Piping / Fittings	Polypropylene plastic tubing, 3/4" ID, 10ft	McMaster	1	\$68.54	\$68.54
					Total Cost:	\$11,440.52

Table 05: Proposed design BOM

Laboratory Filtration Device BOM						
ID	Category	Part Description	Supplier	Qty	Unit Cost (\$ CAD)	Est. Total Cost (\$ CAD)
1	Piping / Fittings	Pipe nipple, 4", 316 SS	McMaster-Carr	1	\$22.94	\$22.94
2	Instrumentation	Pressure Gauge, 1/4"	McMaster-Carr	2	\$22.71	\$45.42
3	Piping / Fittings	pipe nipple, close, 0.75"	McMaster-Carr	3	\$4.90	\$14.69
4	Piping / Fittings	Tee, 0.75" FNPT, 316 SS	McMaster-Carr	1	\$18.99	\$18.99
5	Piping / Fittings	Pipe elbow, 0.75" MNPT,	McMaster-Carr	4	\$6.28	\$25.13
6	Piping / Fittings	Pipe elbow, 0.75" FNPT, PVC	McMaster-Carr	2	\$2.24	\$4.49
7	Treatment	UV Sterilizer, Trojan UV Max	esp Water	1	\$3,642.83	\$3,642.83
8	Piping / Fittings	Tee, 0.75" FNPT, PVC	McMaster-Carr	2	\$4.52	\$9.03
9	Treatment	Purity monitor, hydro-check	Hydro-check	1	\$95.00	\$95.00
10	Treatment	Deionization Resin Tank	Nationwide DI	2	\$479.50	\$959.00
11	Filtration	Sediment filter, 20 GPM,	Grainger	1	\$142.46	\$142.46
12	Piping / Fittings	Polypropylene plastic tubing,	McMaster-Carr	1	\$34.02	\$34.02
					Total Cost:	\$5,014.00

The engineering budget for the original system implementation and the entire redesign cycle is summarized in the table below:

Table 06: Engineering budget report

Engineering Budget Summary					
Original Design					
ID	Description	Notes	Est. hours or quantity	Estimated rate or unit cost (\$ CAD)	Total est. cost (\$ CAD)
1	Construction of system	qualified plumber in BC, Canada installing the system	4	\$32.00	\$128.00
2	Liquid teflon thread sealant		1	\$18.00	\$18.00
3	Design time from original engineer	median value engineer	12	\$42.31	\$507.68
				Total Cost (\$CAD):	\$653.68
Optimized Design					
ID	Description	Notes	Est. hours or quantity	Estimated rate or unit cost (\$ CAD)	Total est. cost (\$ CAD)
4	Engineering analysis (5 team members)	quantifying current parameters, costs, functionality	10	\$42.31	\$423.07
5	Filtration research		4	\$42.31	\$169.23
6	Design calculations		5	\$42.31	\$211.53
7	CAD model development		4	\$42.31	\$169.23
8	Design review and approval		2	\$42.31	\$84.61
9	Construction of system		3	\$32.00	\$96.00
10	Liquid teflon thread sealant		1	\$18.00	\$18.00
				Total Cost (\$CAD):	\$1,171.67
				Combined Integration Cost (\$CAD):	\$1,825.35

The expected incurred costs for implementing the original design consists of the design labour itself and the time for a qualified plumber to install the system. The expected incurred cost in this design cycle was estimated to be approximately \$653.68 CAD. In the next design cycle, the team spent time analyzing the system's functionality, parameters, and cost of components. Once the team had a comprehensive understanding of the systems function, time was spent on researching optimal filtration processes for the laboratory application, performing design calculations, developing a CAD model, and sourcing required components. In addition to the engineering costs, the same integration costs for the installation were assumed, reaching a total design lifecycle cost of \$1,171.67 CAD.

3.2. Work Schedule

The updated project schedule is summarized below as follows:

Table 07: Updated work schedule

Task	Description	Status	Completion Date/ Planned Date
System Inspection	Visited ELW 140 lab and inspected the pump filtration system components	Completed	Thursday, February 26, 2026
System Documentation	Photographed system and identified major components (pump, filters, UV sterilizer, DI tanks, storage tank, gauges, valves)	Completed	Thursday, February 26, 2026
Process Schematic Development	Created a detailed schematic of the system including pipe segments (S0–S8) and fittings (F1–F27)	Completed	Tuesday, March 3, 2026
Pipe Measurements	Measured physical pipe lengths for all segments (S0–S8) in the lab	Completed	Tuesday, March 3, 2026
Fitting Identification	Recorded and catalogued all pipe fittings and their descriptions	Completed	Tuesday, March 3, 2026
System Modeling Framework	Defined pipe segments and assigned fittings to each segment for head loss calculations	Completed	Thursday, March 5, 2026
Determine Pipe Diameters and Roughness	Confirm pipe inner diameters and material roughness values for calculations	Completed	Tuesday, March 10, 2026
Bill of Materials Creation	Compiled a complete bill of materials including pipes, fittings, valves, pump, filters, and instrumentation used in the system	Completed	Tuesday, March 10, 2026
Assign Minor Loss Coefficients	Determine appropriate K values for all fittings using reference tables	Completed	Thursday, March 12, 2026
Major Head Loss Calculations	Calculate friction losses in each pipe segment using equation	Completed	Saturday, March 14, 2026
Minor Loss Calculations	Compute head losses associated with fittings and valves	Completed	Saturday, March 14, 2026
Labor Time Estimation	Estimate the total labor hours required to assemble the system using the bill of materials and installation steps (pipes, fittings, pump, filters, instrumentation)	Completed	Saturday, March 14, 2026
CAD model	Develop a CAD model from common components to visualize function and generate component list	Completed	Sunday, March 15, 2026
Calculation verification	Compare the calculations and CAD model measurements correspond to each other	Completed	Friday, March 20, 2026
New design BOM	Complete and finalize the manufacturing BOM given the finalized CAD model	Completed	Friday, March 20, 2026
System Efficiency Evaluation	Analyze energy use and potential inefficiencies in the system	Completed	Friday, March 20, 2026
Final Results Compilation	Summarize analytical results and conclusions	Completed	Monday, March 23, 2026
Final Report Preparation	Prepare final technical report and presentation materials	Completed	Wednesday, March 25, 2026

4. Concerns and Issues

A significant limitation in this project is the inability to obtain experimental results from the existing system in its current state. The lack of available physical data prevents the team from quantifying the exact water purity achieved with the current filters that were selected. Additionally, testing data could help quantify the exact flow rate contribution from each inlet branch.

Another issue with the current system that led to rudimentary assumptions in the calculations was the variable discharge RO storage tank, which has a continuously reducing working pressure as fluid is pumped from its outlet throughout the piping system. Thus, for the sake of simplicity in the steady-state equations, a constant working pressure for the tank was assumed since the pressure drop over time is not substantial.

5. Recommendations

There are a few ways the system could be improved in the future through experimental validations and monitoring. First, it would be helpful to do actual testing on the system to measure pressure drops across each component, since a lot of the values in this project were based on assumptions, this would help refine the assumptions used in the hydraulic model and improve overall accuracy. Adding pressure gauges before and after major components like the filters, UV sterilization unit, and DI tanks would also help track performance and make it easier to notice when something starts clogging or not working properly. The system should also be flushed if it hasn't been used for a while to prevent bacteria from growing, especially in the deionization tanks. From a design standpoint, putting the system on a movable frame would make it easier to transport between lab spaces and simplify maintenance. Finally, future work could look into selecting different filters or sizes to better balance flow rate, pressure losses, and overall performance depending on how the system is used.

6. Conclusions

In this project, the existing dual-inlet water filtration system was redesigned into a simpler single-inlet system. This started with removing the RO system, as it was not effective for the intended application and added unnecessary complexity. From there, the system was analyzed based on its hydraulic performance, and further improvements were made to simplify the overall design while maintaining functionality.

The original system contained a lot of unnecessary components, including extra fittings, a pump, and a storage tank, which led to high pressure losses and increased energy use. After analyzing different operating cases, it was found that the laboratory faucet already provides enough pressure to run the system, making both the pump and storage tank unnecessary.

The final design simplifies the system to a single inlet, using a combination of a cartridge filter, UV sterilizer, and two deionization tanks, which optimizes the system layout while maintaining good water quality.

From the results, the optimized design significantly reduced losses, with about a 90% reduction in friction losses and an 18% reduction in minor losses. This led to an overall decrease in total head loss and a major improvement in efficiency. In addition, the system

now uses much less energy, requiring only about 33% of the energy compared to the original design to produce the same amount of filtered water. The simplified design also resulted in a cost reduction of approximately 56%, making it much more practical for implementation.

Overall, the new design is simpler, cheaper, and easier to use, while still meeting the needs of the lab. This project shows how removing unnecessary components and improving system layout can have a big impact on performance and efficiency.

7. References

- [1] F. M. White, *Fluid Mechanics*, 8th ed. New York, NY: McGraw-Hill Education, 2016.
- [2] P. G, “Can water filtration systems reduce water pressure in my home?,” *The Clean Well Water Report*, Dec. 09, 2025. https://www.cleanwaterstore.com/blog/can-water-filtration-systems-reduce-water-pressure-in-my-home/?srsltid=AfmBOopmj9U_tYdxpnWKyVesZxNI9H2rylYP7Owx_0IeqcjgPPJkC4xy (accessed Mar. 25, 2026).
- [3] “DI Tank User Guide.” Available: <https://puretecwater.com/wp-content/uploads/2024/05/di-tank-user-guide-1.pdf>
- [4] Sachin, “Deionized Water System vs Reverse Osmosis: Which is Best For Your Application?,” *Reynolds Culligan*, Sep. 18, 2020. <https://www.reynoldsculligan.com/deionize-d-water-system-vs-reverse-osmosis/>
- [5] J. Nriagu, C. Xi, A. Siddique, A. Vincent, and B. Shomar, “Influence of Household Water Filters on Bacteria Growth and Trace Metals in Tap Water of Doha, Qatar,” *Scientific Reports*, vol. 8, no. 1, May 2018, doi: <https://doi.org/10.1038/s41598-018-26529-8>.

Appendix A: Original System Calculations

Hydraulic Model Assumptions / Controls					
Section	Parameter	Value	Units	Use in model	Source / rationale
Fluid	Density, ρ	1000.000	kg/m ³	All cases	Water
Fluid	Dynamic viscosity, μ	0.001	Pa-s	All cases	Water near room temperature
Fluid	Gravity, g	9.810	m/s ²	All cases	Standard
Geometry	Main tubing ID	0.01524	m	S0-S7	Diagrams sheet note: 0.6 ID, 0.75 OD polypropylene
Geometry	Main tubing roughness	0.000	m	S0-S7	Polypropylene roughness from original workbook
Geometry	Tank branch 1 in ID assumption	0.0266446	m	S8 1 in portion	Approx. 1 in Sch40 ID / fitting bore; verify physically
Geometry	Tank branch total length	0.55	m	S8	Measured in lab
Geometry	Fraction of S8 treated as 1 in	0.500	-	S8	Assumption because measured split between 1 in and 3/4 in hardware was not recorded
Supply	Tap/faucet pressure	344.7378647	kPa	Cases with tap	Original workbook assumption
Supply	Tank gauge pressure	275.7902917	kPa	Cases with tank	Original workbook assumption
Filter	Rated filter ΔP	13789.51459	Pa	Filter loss	Original workbook note: approx. 2 psi at 2 gpm
Filter	Rated filter flow	0.00012618	m ³ /s	Filter loss scaling	Used for $\Delta P \sim (Q/Q_{\text{rated}})^2$ scaling
UV	UV equivalent length	0.9144	m	Downstream loss	
DI	DI vessel rated ΔP	34473.78647	Pa	Each DI tank	
DI	DI vessel rated flow	0.000189271	m ³ /s	Each DI tank	
Fittings	K, 0.75 in elbow	1.750	-	F2/F8/F12/etc.	Original workbook / table
Fittings	K, 1 in elbow	1.500	-	F25	Original workbook / table
Fittings	K, tee line flow	0.900	-	Tee fittings	Original workbook / table
Fittings	K, ball valve fully open	0.050	-	F26	Original workbook / table
Fittings	K, pressure regulating valve	1.560	-	PRV1	Original workbook / table, 20° closed
Fittings	K, bushing / reducer default	0.200	-	Reducer/bushing fittings	
Fittings	K, threaded nipple default	0.100	-	F1	
Elevation	Static rise tee/pump level to outlet	0.4699	m	Pump requirement	Original workbook value

Figure A1: Original system assumptions

Hydraulic Calculations						
		Case 1: Tank + Stillwater	Case 2: Tank + Tap	Case 3: Stillwater only	Case 4: Tank only	Case 5: Tap only
Section	Variable / units					
Case Inputs						
	Tap pressure at filter inlet [kPa]	0	344.7378647	0	0	344.7378647
	Tank pressure at storage branch [kPa]	275.7902917	275.7902917	0	275.7902917	0
	Filter-side flow Q1 [m ³ /s]	0.000315451	0.000189271	0.000252361	0	0.000189271
	Tank-side flow Q2 [m ³ /s]	3.15451E-05	3.15451E-05	0	3.15451E-05	0
	Total flow Qt [m ³ /s]	0.000346996	0.000220816	0.000252361	3.15451E-05	0.000189271
Geometry / Fluid Properties						
	Density, ρ [kg/m ³]	1000	1000	1000	1000	1000
	Dynamic viscosity, μ [Pa·s]	0.001002	0.001002	0.001002	0.001002	0.001002
	Gravity, g [m/s ²]	9.81	9.81	9.81	9.81	9.81
	Main line ID, Dmain [m]	0.01524	0.01524	0.01524	0.01524	0.01524
	Main line area, Amain [m ²]	0.000182415	0.000182415	0.000182415	0.000182415	0.000182415
	Main line roughness, ϵ [m]	0.0000015	0.0000015	0.0000015	0.0000015	0.0000015
	S8 total length [m]	0.55	0.55	0.55	0.55	0.55
	S8 fraction at 1 in [-]	0.5	0.5	0.5	0.5	0.5
	S8 1 in ID, D8a [m]	0.0266446	0.0266446	0.0266446	0.0266446	0.0266446
	S8 0.75 in ID, D8b [m]	0.01524	0.01524	0.01524	0.01524	0.01524
	S8 1 in length, L8a [m]	0.275	0.275	0.275	0.275	0.275
	S8 0.75 in length, L8b [m]	0.275	0.275	0.275	0.275	0.275
Upstream Filter Branch (Inlet to Tee)						
	Lfilter = S0+S1+S2 [m]	1.67	1.67	1.67	1.67	1.67
	Vfilter [m/s]	1.729306876	1.037584126	1.383445501	0	1.037584126
	Refilter [-]	26302.03273	15781.21964	21041.62618	0	15781.21964
	f _{filter} [-]	0.024408637	0.027639434	0.025740104	0	0.027639434
	Filter ΔP actual [Pa]	86184.46616	31026.40782	55158.05835	0	31026.40782
	Filter ΔP head [m]	8.78536862	3.162732703	5.622635917	0	3.162732703
	K upstream minors [-]	3.55	3.55	3.55	3.55	3.55
	Upstream major loss hf _{filter} [m]	0.407680714	0.1661913	0.275148347	0	0.1661913
	Upstream minor loss hm _{filter} [m]	0.541094957	0.194794185	0.346300773	0	0.194794185
	Pressure head available at tee from filter branch [m]	-9.734144291	31.61775629	-6.244085036	0	31.61775629
Storage Branch (Tank to Tee)						
	Area 1 in branch, A8a [m ²]	0.000557581	0.000557581	0.000557581	0.000557581	0.000557581
	Area 0.75 in branch, A8b [m ²]	0.000182415	0.000182415	0.000182415	0.000182415	0.000182415
	V8a [m/s]	0.056574874	0.056574874	0	0.056574874	0
	V8b [m/s]	0.172930688	0.172930688	0	0.172930688	0
	Re8a [-]	1504.406066	1504.406066	0	1504.406066	0
	Re8b [-]	2630.203273	2630.203273	0	2630.203273	0
	f8a [-]	0.042541706	0.042541706	0	0.042541706	0
	f8b [-]	0.049284625	0.049284625	0	0.049284625	0
	Tank branch major loss hf8 [m]	0.001427144	0.001427144	0	0.001427144	0
	K tank branch minors [-]	4.21	7.21	4.21	4.21	4.21
	Tank branch minor loss hm8 [m]	0.006416929	0.010989562	0	0.006416929	0
	Pressure head available at tee from tank branch [m]	28.10533551	28.10076288	0	28.10533551	0

Figure A2: Original system calculations for all five operating cases 1/2

Hydraulic Calculations						
		Case 1: Tank + Stillwater	Case 2: Tank + Tap	Case 3: Stillwater only	Case 4: Tank only	Case 5: Tap only
Section	Variable / units					
	Tank branch minor loss h_{m8} [m]	0.006416929	0.010989562	0	0.006416929	0
	Pressure head available at tee from tank branch [m]	28.10533551	28.10076288	0	28.10533551	0
Tee Mixing Approximation and Downstream Treatment Line						
	Flow-weighted tee pressure head [m]	2.555030501	31.11532866	0	28.10533551	31.61775629
	Ldown = S3+S4+S5+S6+S7 [m]	3.365	3.365	3.365	3.365	3.365
	Qt velocity in downstream line [m/s]	1.902237564	1.210514813	1.383445501	0.172930688	1.037584126
	Redown [-]	28932.236	18411.42291	21041.62618	2630.203273	15781.21964
	fdown [-]	0.023874309	0.026594651	0.025740104	0.046581043	0.027639434
	Downstream straight-pipe h_f [m]	0.972	0.439	0.554	0.016	0.335
	UV equivalent-length loss [m]	0.264187681	5.622636086	0.150656077	0.004259961	0.090997201
	Downstream fitting K total [-]	10.400	10.400	10.400	10.400	10.400
	Downstream fitting loss h_m [m]	1.9181	0.7767	1.0145	0.0159	0.5707
	Each DI vessel ΔP_{actual} [Pa]	115870.2267	46922.6538	61286.73149	957.6051796	34473.78647
	Total DI head loss (2 vessels) [m]	23.62288007	9.566290275	12.49474648	0.195230414	7.028294896
	Static rise to outlet [m]	0.4699	0.4699	0.4699	0.4699	0.4699
	Required pump head h_p [m]	24.692	-14.241	14.684	-27.404	-23.123
	Pump required? [text]	Yes	No	Yes	No	No
	Assumed pump efficiency [-]	0.7	0	0.7	0	0
	Pump shaft power [W]	120.0758392	0	51.93308362	0	0
	Pump power [HP]	0.161024353	0	0.069643412	0	0

Figure A3: Original system calculations for all five operating cases 2/2

Table 6.5 Resistance Coefficients $K = h_m/[V^2/(2g)]$ for Open Valves, Elbows, and Tees		Nominal diameter, in								
		Screwed				Flanged				
		$\frac{1}{2}$	1	2	4	1	2	4	8	20
Valves (fully open):										
Globe	14	8.2	6.9	5.7	13	8.5	6.0	5.8	5.5	
Gate	0.30	0.24	0.16	0.11	0.80	0.35	0.16	0.07	0.03	
Swing check	5.1	2.9	2.1	2.0	2.0	2.0	2.0	2.0	2.0	
Angle	9.0	4.7	2.0	1.0	4.5	2.4	2.0	2.0	2.0	
Elbows:										
45° regular	0.39	0.32	0.30	0.29						
45° long radius					0.21	0.20	0.19	0.16	0.14	
90° regular	2.0	1.5	0.95	0.64	0.50	0.39	0.30	0.26	0.21	
90° long radius	1.0	0.72	0.41	0.23	0.40	0.30	0.19	0.15	0.10	
180° regular	2.0	1.5	0.95	0.64	0.41	0.35	0.30	0.25	0.20	
180° long radius					0.40	0.30	0.21	0.15	0.10	
Tees:										
Line flow	0.90	0.90	0.90	0.90	0.24	0.19	0.14	0.10	0.07	
Branch flow	2.4	1.8	1.4	1.1	1.0	0.80	0.64	0.58	0.41	

Figure A4: Minor loss coefficients used in calculations [1]

ϵ				
Material	Condition	ft	mm	Uncertainty, %
Steel	Sheet metal, new	0.00016	0.05	±60
	Stainless, new	0.000007	0.002	±50
	Commercial, new	0.00015	0.046	±30
	Riveted	0.01	3.0	±70
	Rusted	0.007	2.0	±50
Iron	Cast, new	0.00085	0.26	±50
	Wrought, new	0.00015	0.046	±20
	Galvanized, new	0.0005	0.15	±40
	Asphalted cast	0.0004	0.12	±50
Brass	Drawn, new	0.000007	0.002	±50
Plastic	Drawn tubing	0.000005	0.0015	±60
Glass	—	Smooth	Smooth	
Concrete	Smoothed	0.00013	0.04	±60
	Rough	0.007	2.0	±50
Rubber	Smoothed	0.000033	0.01	±60
Wood	Stave	0.0016	0.5	±40

Figure A5: Absolute roughness values used in calculations [1]

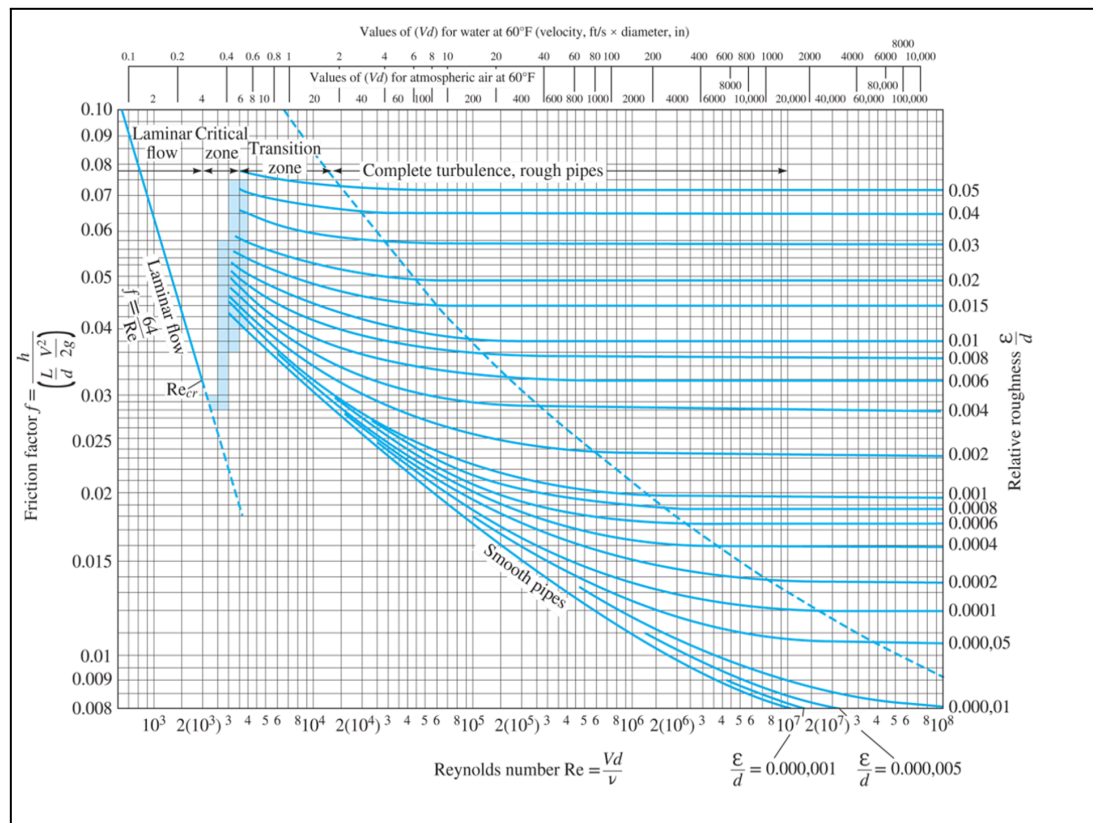


Figure A6: Moody chart used in calculations [1]

Appendix B: Proposed System Design Calculations

Hydraulic Calculations				
Section	Variable / units	Case 5: Tap only		Notes
Case Inputs				
	Tap pressure at filter inlet [kPa]	344.7378647		
	Head Provided by Tap [m]	35.14147448		
	System flow Q [m ³ /s]	0.000189271		Flow from tap (3 GPM) in m ³ /s
Geometry / Fluid Properties				
	Density, ρ [kg/m ³]	1000		
	Dynamic viscosity, μ [Pa·s]	0.001002		
	Gravity, g [m/s ²]	9.81		
	Main line ID, D_{main} [m]	0.01524		0.75" PVC pipe
	Main line area, A_{main} [m ²]	0.000182415		
	Main line roughness, ϵ [m]	0.0000015		Smooth PVC pipe
	S8 total length [m]	0.55		
	S8 fraction at 1 in [-]	0.5		
	S8 1 in ID, D_{8a} [m]	0.0266446		
	S8 0.75 in ID, D_{8b} [m]	0.01524		
Filter Branch (Tap inlet elbow to outlet before UV elbow)				
	L_1	0.1016		length: 4"
	V_1 [m/s]	1.037584126		
	Re_1 [-]	15781.21964		
	f_1 [-]	0.027639434		
	Filter ΔP actual [Pa]	31026.40782		scale ΔP with $(Q/Q_{rated})^2$
	Filter ΔP head [m]	3.162732703		$P/\rho g$
	Upstream minor loss coefficients [-]	1.95		upstream filter branch has: 2 nipple (0.75"), one elbow from inlet (0.75")
	Upstream major loss h_{f1} [m]	0.0101108		$h_f = f \cdot (L/D) \cdot V^2/(2g)$
	Upstream minor loss h_{m1} [m]	0.106999623		$h_m = K \cdot V^2/(2g)$
	Pressure head available at UV inlet [m]	31.86163136		supply_head - friction - minors - filter_head
UV Branch (UV inlet elbow to outlet)				
	Area 0.75 in branch, A_2 [m ²]	0.000182415		
	V_2 [m/s]	1.037584126		
	Re_2 [-]	15781.21964		
	f_2 [-]	0.027639434		Using Swamee-Jin Equation
	Upstream major loss h_{f2} [m]	0.0101108		length: 4"
	UV Filter loss h_{UV} [m]	5.622636086		Head loss from filter based on 40% of rated flow which is a pressure drop of 4psi(reference: chrome-extension://efaidnbmninnipcbajpcglclefindmkaj/https://www.home-water-purifiers-and-filters.com/manuals/Engineering-UVMax-Pro-Series.pdf
	Upstream minor loss coefficients [-]	1.75		elbow into UV filter (0.75")
	Upstream minor loss h_{m2} [m]	0.096025302		
	Pressure head available at UV outlet [m]	26.13285917		

Figure B1: Optimized system design calculations 1/2

Hydraulic Calculations				
Section	Variable / units	Case 5: Tap only		Notes
UV Branch (UV inlet elbow to outlet)				
	Area 0.75 in branch, A_2 [m ²]	0.000182415		
	V_2 [m/s]	1.037584126		
	Re_2 [-]	15781.21964		
	f_2 [-]	0.027639434		Finally, the Swamee-Jain Equation is $f = \frac{0.250}{\left[\log \left[\frac{5.74}{Re^{0.9}} + \frac{5.74}{Re^{1.75}} \right] \right]^2}$
	Upstream major loss h_{f2} [m]	0.0101108		Using Swamee-Jin Equation length: 4"
	UV Filter loss h_{UV} [m]	5.622636086		Head loss from filter based on 40% of rated flow which is a pressure drop of 4psi(reference: chrome-extension://efaidnbmnnnibpcjpcglclefindmkaj/https://www.home-water-purifiers-and-filters.com/manuals/Engineering-UVMMax-Pro-Series.pdf
	Upstream minor loss coefficients [-]	1.75		elbow into UV filter (0.75")
	Upstream minor loss h_{m2} [m]	0.096025302		
	Pressure head available at UV outlet [m]	26.13285917		
Deionization tank Branch				
	Area 0.75 in branch, A_3 [m ²]	0.000182415		
	Velocity V_3 [m/s]	1.037584126		
	Re_3 [-]	15781.21964		
	f_3 [-]	0.032916685		
	Upstream Major Loss h_{f3} [m]	0.04148077		Total Pipe segment length 14"
	Deionization Tank head loss h_{tank} [m]	7.028		
	Upstream minor loss coefficients [-]	3.250		6 bushings, 3 tees, 1 elbow from UV filter
	Upstream minor losses h_{m3} [m]	0.17833270		
	Pressure head available at system outlet [m]	18.8848		
	Total System Head Losses [m]	16.25672		
	Initial Total System Loss [m]	20.44160		
	Total Head Saved [m]	4.18488		
	Total Friction Losses	0.06170		
	Total Minor Losses	16.19502		
	Initial Friction Losses	0.63752		
	Initial Minor Losses	19.80408		

Figure B2: Optimized system design calculations 2/2